An Internship Report on

PYTHON DEVELOPMENT INTERNSHIP

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: CS-2341460

Name of Student: NEHA SINHA

Semester: 7th Semester   Section: 4CSE12

Batch: 2023-2027

CANDIDATE'S DECLARATION

I, Neha Sinha, do hereby declare that the internship report titled "Python Development Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: NEHA SINHA
Roll No.: CS-2341460

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Squarcell Resource India Pvt. Ltd. (QSkill) for providing me the opportunity to undertake this virtual internship in Python Development, and for their guidance throughout its duration.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: NEHA SINHA
Roll No.: CS-2341460

TABLE OF CONTENTS

CHAPTER 3: INTERNSHIP COMPLETION CERTIFICATE

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report documents the work completed during a one-month virtual internship in Python Development undertaken with QSkill, a program run by Squarcell Resource India Pvt. Ltd. (SR India). The internship focused on strengthening core Python programming skills through three hands-on projects: data analysis with pandas, a matrix operations tool built with NumPy, and a machine learning model for house price prediction using scikit-learn. Each project was designed to reinforce a distinct area of Python's ecosystem — data handling, numerical computation, and predictive modelling — while following good coding and documentation practices.

4.2 Organization Profile

Squarcell Resource India Pvt. Ltd., operating under the brand SR India / QSkill, is an ISO 9001:2015 certified organization based in Navi Mumbai, Maharashtra, that provides virtual internship and upskilling programs across various technology domains. QSkill's internship programs are designed to give students hands-on, project-based learning experience with flexible timing, aimed at building practical, job-ready skills alongside academic study.

4.3 Problem Statement

Students often learn Python programming concepts in isolation without applying them to realistic, end-to-end problems. This internship addressed that gap by requiring the design, development, and documentation of three independent but progressively complex Python projects — covering data analysis, numerical/matrix computation, and applied machine learning — each solving a self-contained, practical problem from start to finish.

4.4 Project Objectives

- To analyze a structured dataset using pandas and derive meaningful statistical insights.
- To build a reusable, menu-driven command-line tool for common matrix operations using NumPy.

- To train and evaluate a linear regression model using scikit-learn to predict a continuous target variable.
- To practice reading, cleaning, and visualizing data, and to present results and findings clearly through written reports.
- To gain experience with version control and hosting completed work on GitHub.

4.5 Scope of the Project

The scope of the internship work covered three self-contained projects rather than a single large application, reflecting the structured task-based format of the internship programme:

- Data Analysis Task — exploratory analysis of a student examination score dataset.
- Matrix Operations Tool — a command-line calculator supporting core linear algebra operations.
- Linear Regression — House Price Prediction — a supervised machine learning model trained on property features to estimate house prices.

Each project was scoped to be completed independently, with its own dataset (or user input), source code, output artifacts, and documentation, and all three were version-controlled and published on GitHub.

4.6 Technologies and Tools Used

- Programming Language: Python 3
- Libraries: pandas, NumPy, matplotlib, scikit-learn
- Development Environment: Visual Studio Code (VS Code)
- Version Control & Hosting: Git and GitHub
- Documentation: Markdown (README files) and Microsoft Word (Report)

4.7 System Architecture

Each project follows a simple, linear data-flow architecture suited to a script-based Python application:

- Input Layer: a CSV dataset (Data Analysis, Linear Regression) or direct user input via the terminal (Matrix Operations Tool).
- Processing Layer: pandas/NumPy/scikit-learn functions perform cleaning, computation, model training, or matrix arithmetic.
- Output Layer: results are printed to the console, saved as processed CSV files, and visualized using matplotlib charts saved as PNG images.

This straightforward architecture keeps each tool easy to run, test, and extend, and mirrors how small, task-focused Python scripts are typically structured in practice.

4.8 Methodology

For the Data Analysis Task, a dataset of student scores across five subjects was loaded into a pandas DataFrame, checked for missing values, and summarized using descriptive statistics (mean, standard deviation, minimum, and maximum). A new column was derived to compute each student's average score, and students were ranked accordingly. Results were visualized using matplotlib bar charts.

For the Matrix Operations Tool, a menu-driven command-line interface was built using NumPy to perform addition, subtraction, multiplication, scalar multiplication, transpose, determinant, and inverse operations on user-entered matrices, with input validation and clear error handling for invalid dimensions or singular matrices.

For the Linear Regression project, a synthetic dataset of house features (area, bedrooms, bathrooms, age, and distance from the city centre) was used to train a scikit-learn LinearRegression model. The data was split into training (80%) and test (20%) sets; the model was trained on the training set and evaluated on the held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score.

4.9 Expected Outcomes

The Data Analysis Task produced subject-wise summary statistics, a ranked list of students by average score, and comparative bar charts of subject and student performance.

The Matrix Operations Tool produced a fully working, error-handled command-line calculator capable of performing all seven core matrix operations correctly on user-supplied matrices of any valid dimension.

The Linear Regression project produced a trained model achieving an R^2 score of 0.9966 on the test set, with an average prediction error (MAE) of approximately Rs. 1,30,000, along with a chart comparing actual versus predicted prices and a breakdown of how each feature influences the predicted price.

All three projects, along with their source code, datasets, output files, and README documentation, were pushed to a public GitHub repository, organized into separate folders per project.

4.10 Certificates of Completion and Other Communication Proof

The Offer Letter received from Squarcell Resource India Pvt. Ltd. (QSkill), confirming selection for the Python Development internship, is attached below as proof of communication.

SR INDIA

QSkill

To:
Neha Sinha

July 22nd, 2026

Congratulations on your selection for the Python Development internship at **QSkill.** The 1-month internship runs from 25th July to 25th August 2026, focusing on learning, skill development, and hands-on experience. You are expected to perform tasks diligently and follow instructions, with the aim of enhancing your career growth. Best wishes for a successful internship!

Details:

- **Mode:** Virtual
- **Timing:** Flexible
- **Duration:** 1 month
- **Stipend:** Unpaid

For, **Squarcell Resource India Pvt. Ltd**

Authorised Signatory

https://www.qskill.in Navi Mumbai, Maharashtra career@srindia.co

The Internship Completion Certificate issued upon completion of the internship has been attached earlier in this report under Chapter 3.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

1. pandas documentation — https://pandas.pydata.org/docs/

2. NumPy documentation — https://numpy.org/doc/

3. matplotlib documentation — https://matplotlib.org/stable/

4. scikit-learn documentation — https://scikit-learn.org/stable/

5. Python 3 official documentation — https://docs.python.org/3/

6. QSkill / Squarcell Resource India Pvt. Ltd. — https://www.qskill.in

7. Project source code repository — https://github.com/Nehailm/-2023-2027_Neha_Sinha_CS-2341460_7thSem_4CSE12